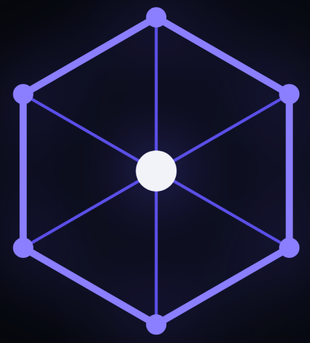


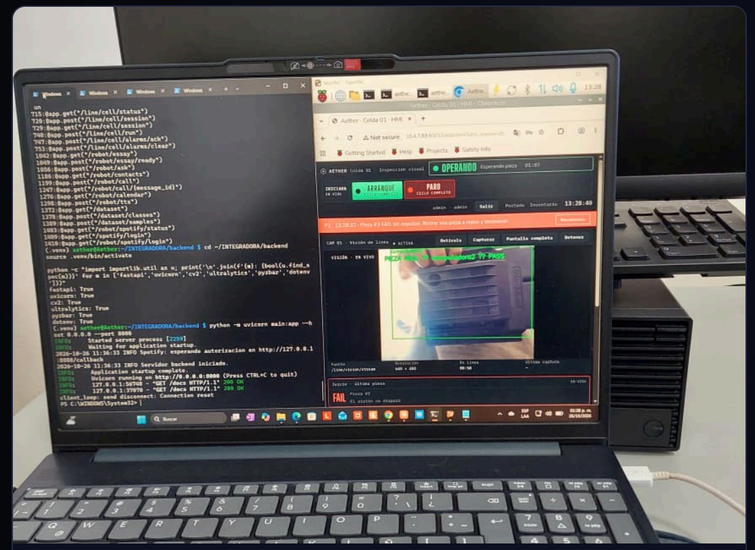
AETHER

INDUSTRIAL SYSTEMS

Mario Iván Méndez Álvarez · Uriel Armando Rivero Rivero · Juan Emmanuel Martínez Aguirre · Pedro José Pallares Muñoz · David Gerardo Garnica Chaparro · Sadher Isaac Torres Galindo

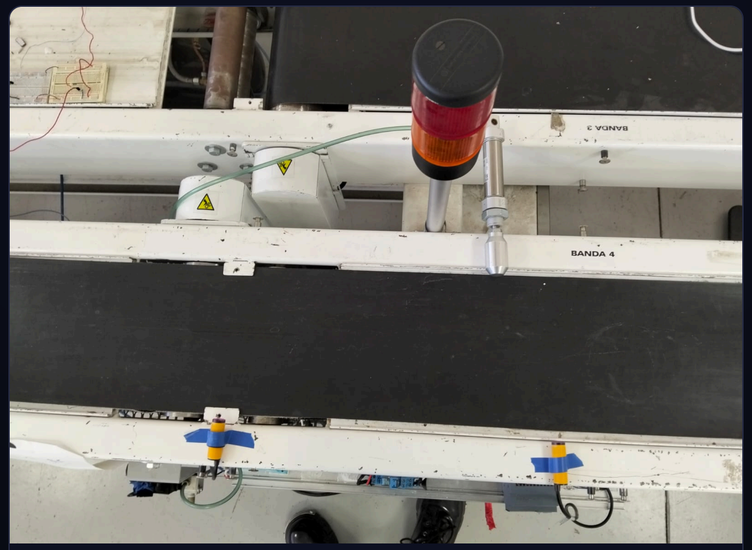
INTRODUCTION

A small manufacturing line that does three jobs usually solved apart



Inspect

Computer vision checks every part on the belt.



Eject

A piston removes the part when it is scrap.



Transport

A mobile robot carries good parts and keeps a traceable record.

Objective

Develop a working automotive manufacturing line that integrates a conveyor with positioning fixtures, a vision check, a scrap-ejecting piston and a mobile inventory unit, running the full cycle of positioning, verification and transport.

ABSTRACT

The project in three steps

Need

Small lines depend on operators to move parts, verify assembly and count stock: downtime, uneven quality, counting errors.

Method

Architecture decisions closed first, then modules built and tested separately before integrating.

Result

Cell software, inventory engine, server and dashboard run with automated tests. The robot is the remaining critical path.

JUSTIFICATION

Why a third way

Complete commercial systems

Usually exceed the budget of a small or medium company.

Partial solutions

Leave coordination gaps that end up being solved by hand.

Aether

Affordable integration, plus a data layer that never silently overwrites a result.

COMPARISON

Without our system vs. with our system

	WITHOUT	WITH AETHER
Moving parts	Operators between stations	Mobile robot
Quality check	Judgment varies by person and shift	Vision: 5 equal readings in a row
Scrap	Removed by hand	Ejected by piston
Coordination	The operator is the link: downtime	Direct handshake cell to cart
Inventory	Manual count, errors add up	Observed vs. declared, kept apart
Traceability	Scattered or missing	Full log of every run

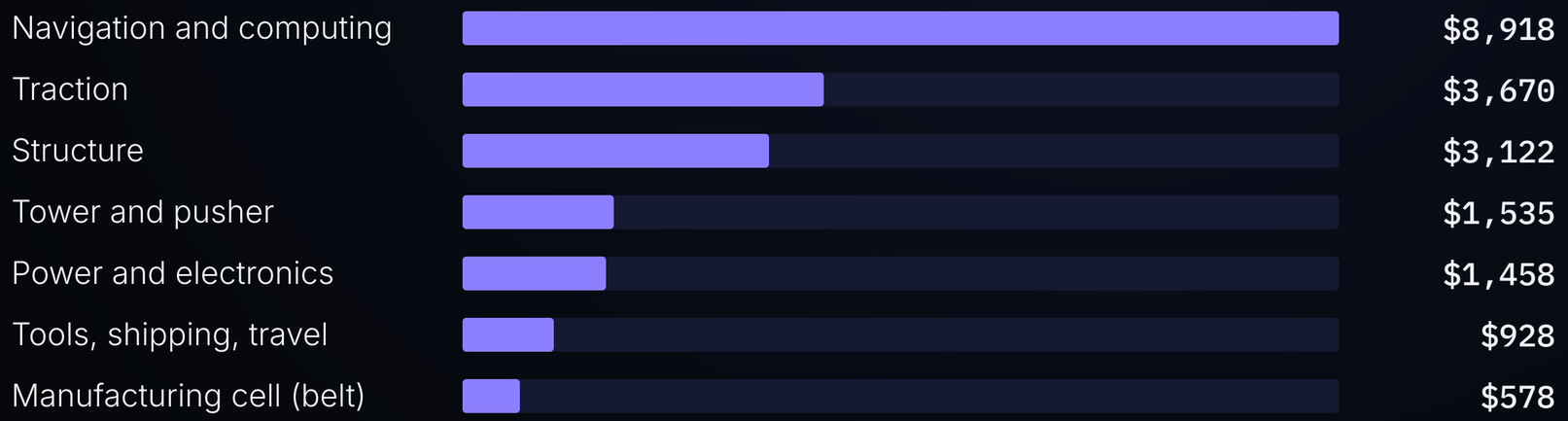
SCHEDULE

Activity schedule



PRICING

What we spent, by subsystem



PAID SO FAR

$20,208

MXN

PROJECTED MATERIALS

$23,703 – $26,189

MXN

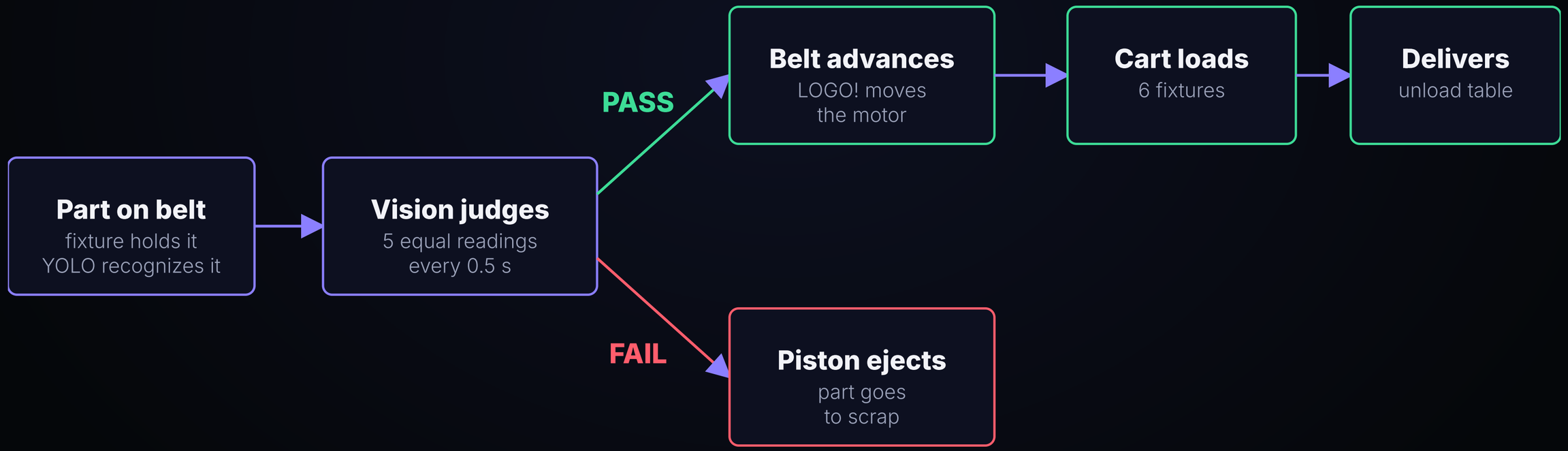
PRICING

Quotation for a final user

ITEM	BASIS	MXN
Materials	Projected purchase list (upper bound)	26,189
Labor	6 people × 240 h × $75/h (estimate)	108,000
Build time	14 weeks	—
Technologies	LOGO! PLC, ESP32, Raspberry Pi 4, Jetson Orin Nano, YOLO, FastAPI	—
Final-user price	**Materials + labor + 15% margin (estimate)**	≈ 154,000

WORKFLOW

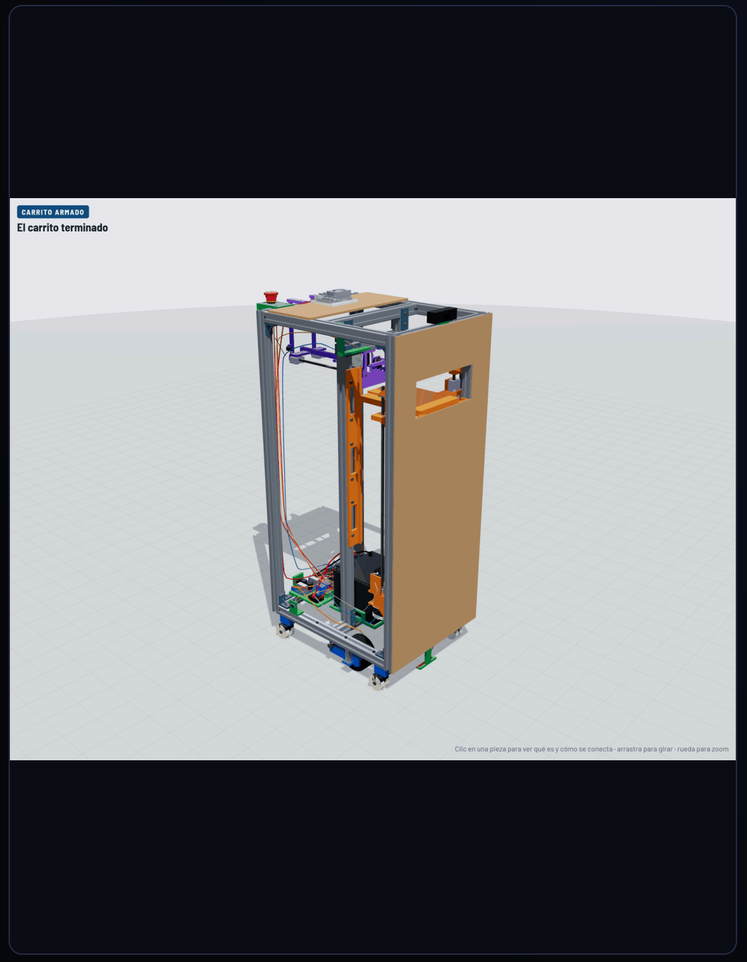
General workflow



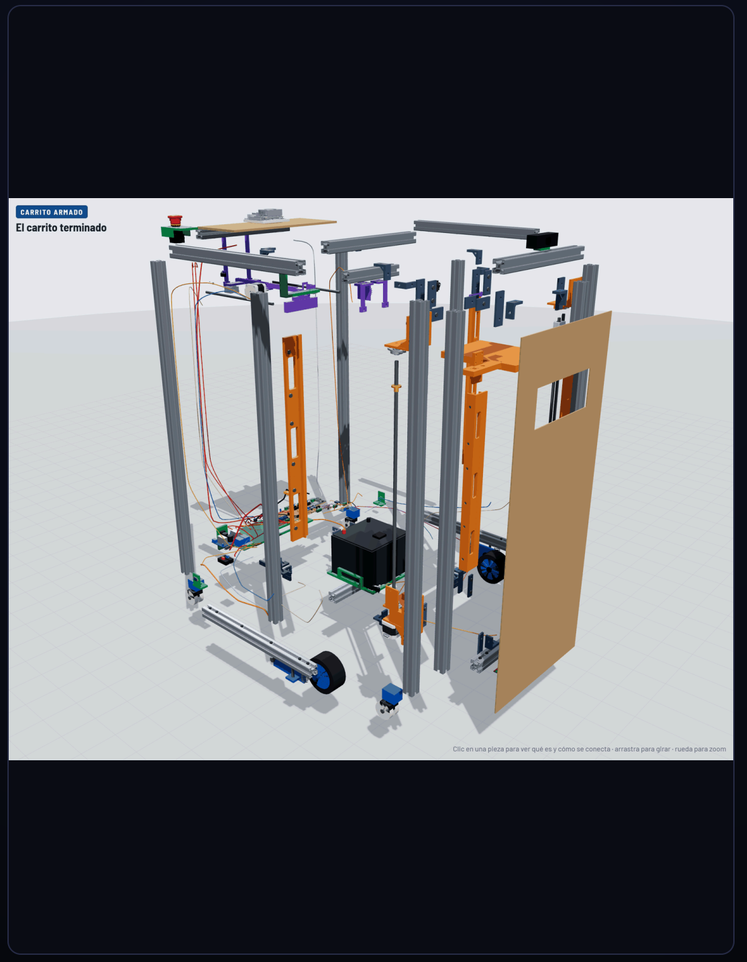
Every step is logged on the server: observed data stays separate from declared inventory

CONSTRUCTION · STAGE 1

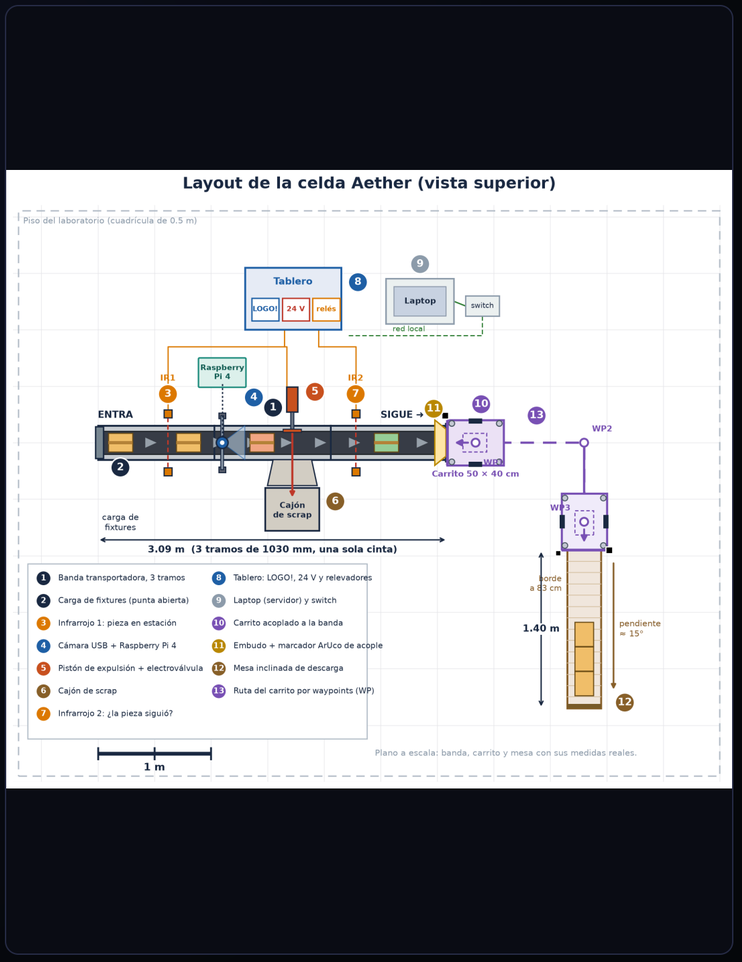
Design first: every part in 3D



Assembled cart, 3D build sheet



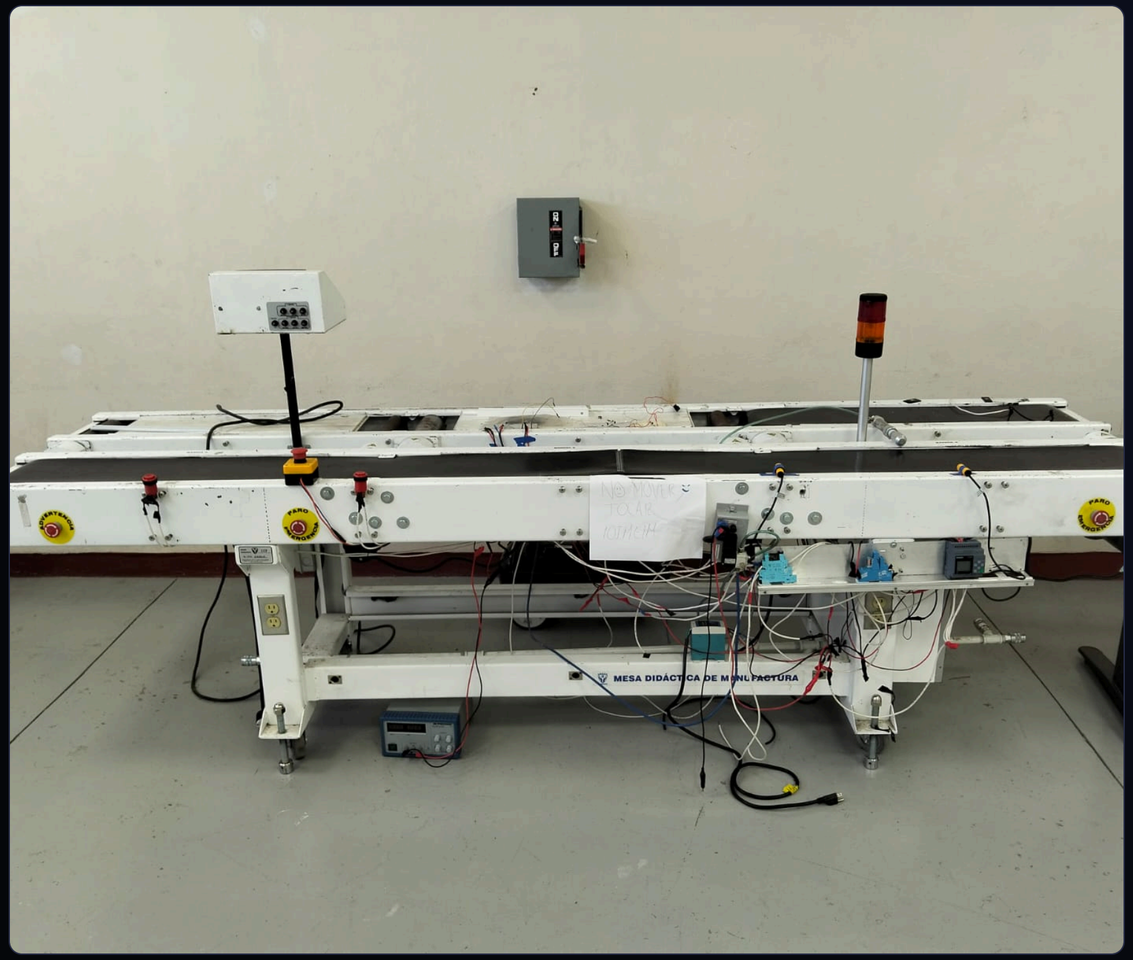
Exploded view of every part



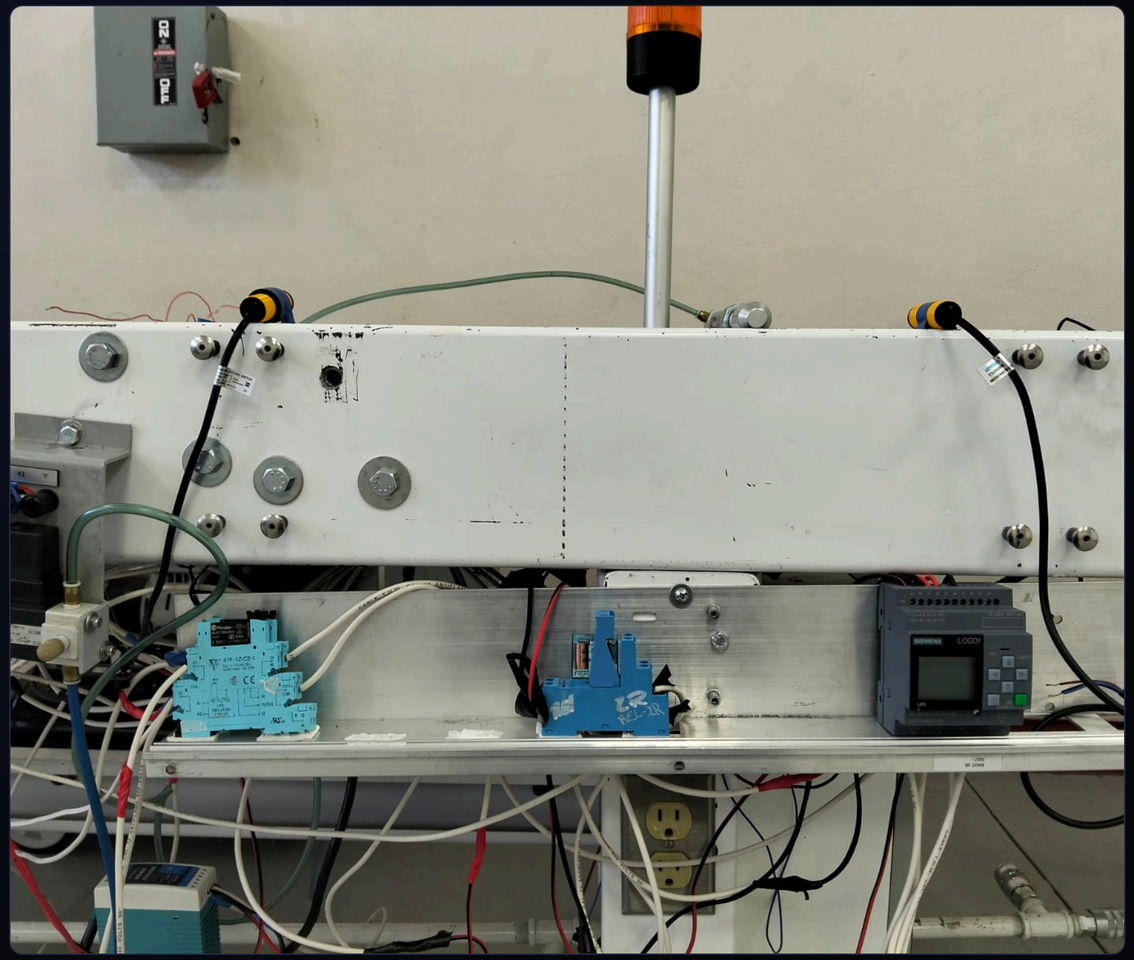
Layout: cell, cart and unload table

CONSTRUCTION · STAGE 2

The manufacturing cell, built

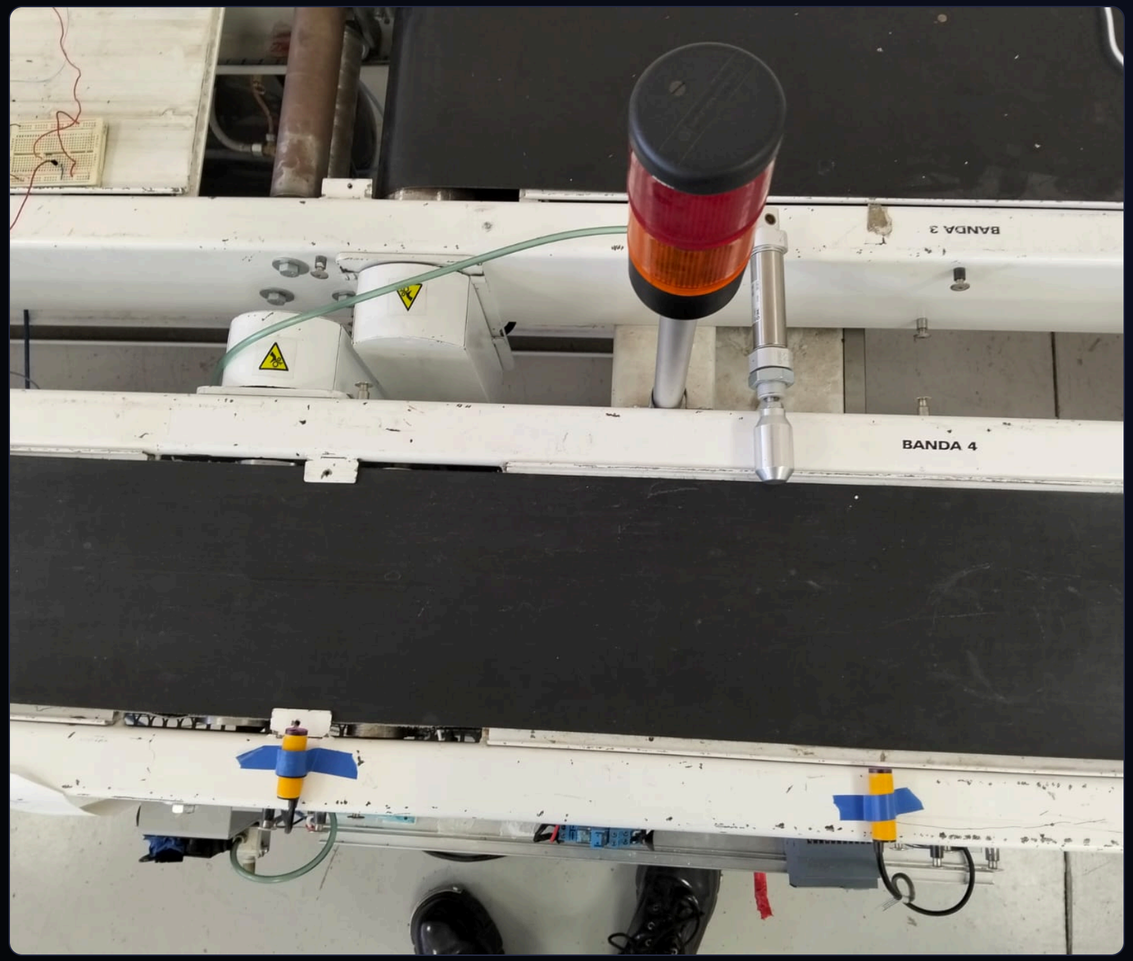


General view of the assembled cell

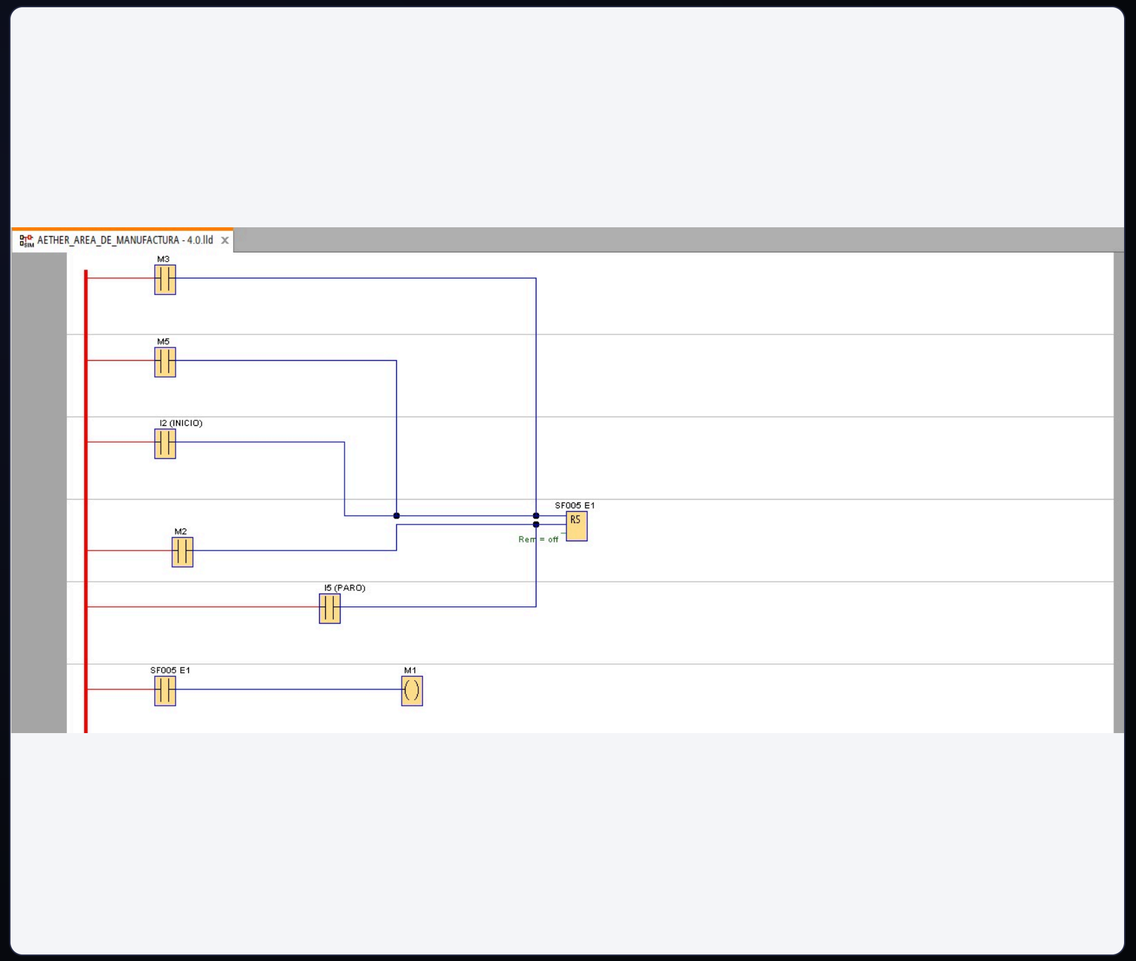


Panel: LOGO!, 24 V supply and relays

Pneumatics: the scrap piston



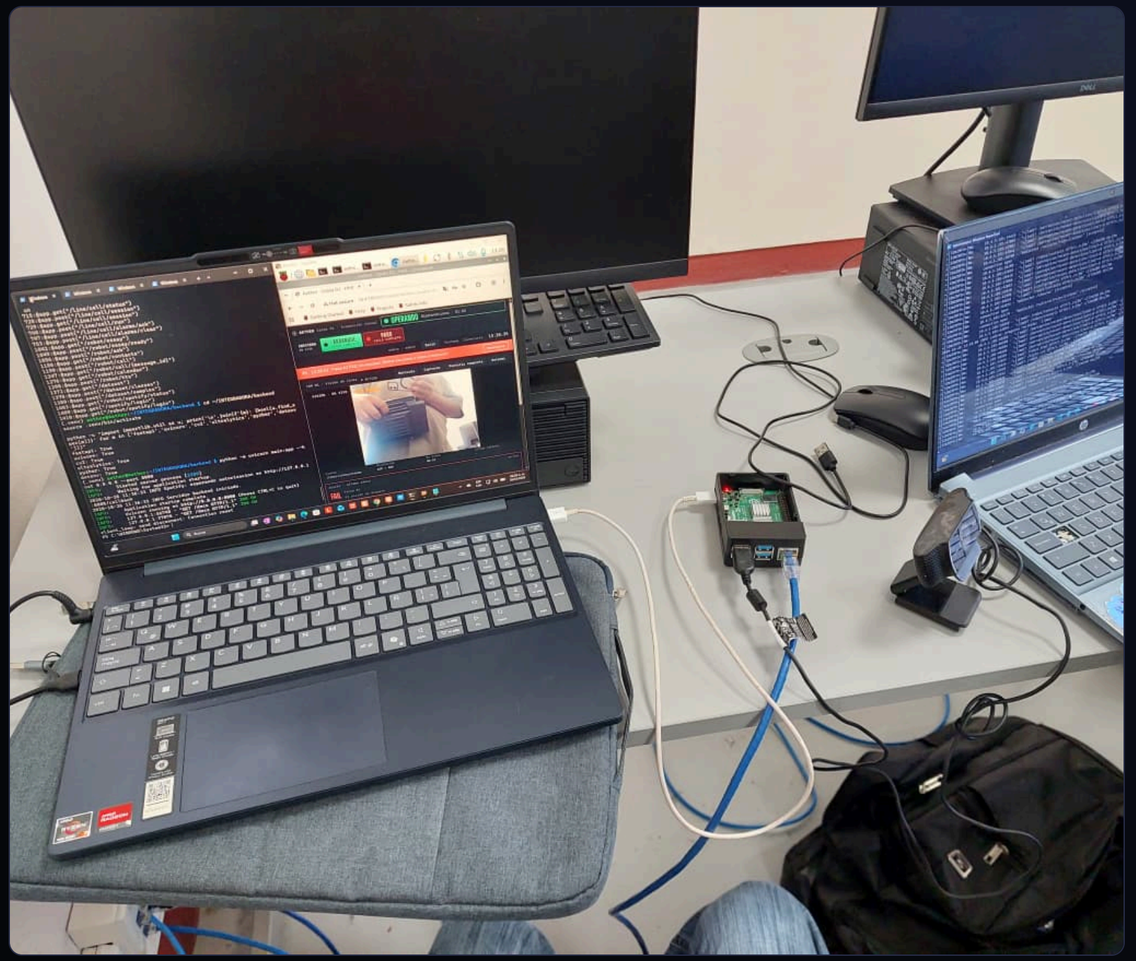
Scrap piston and its solenoid valve



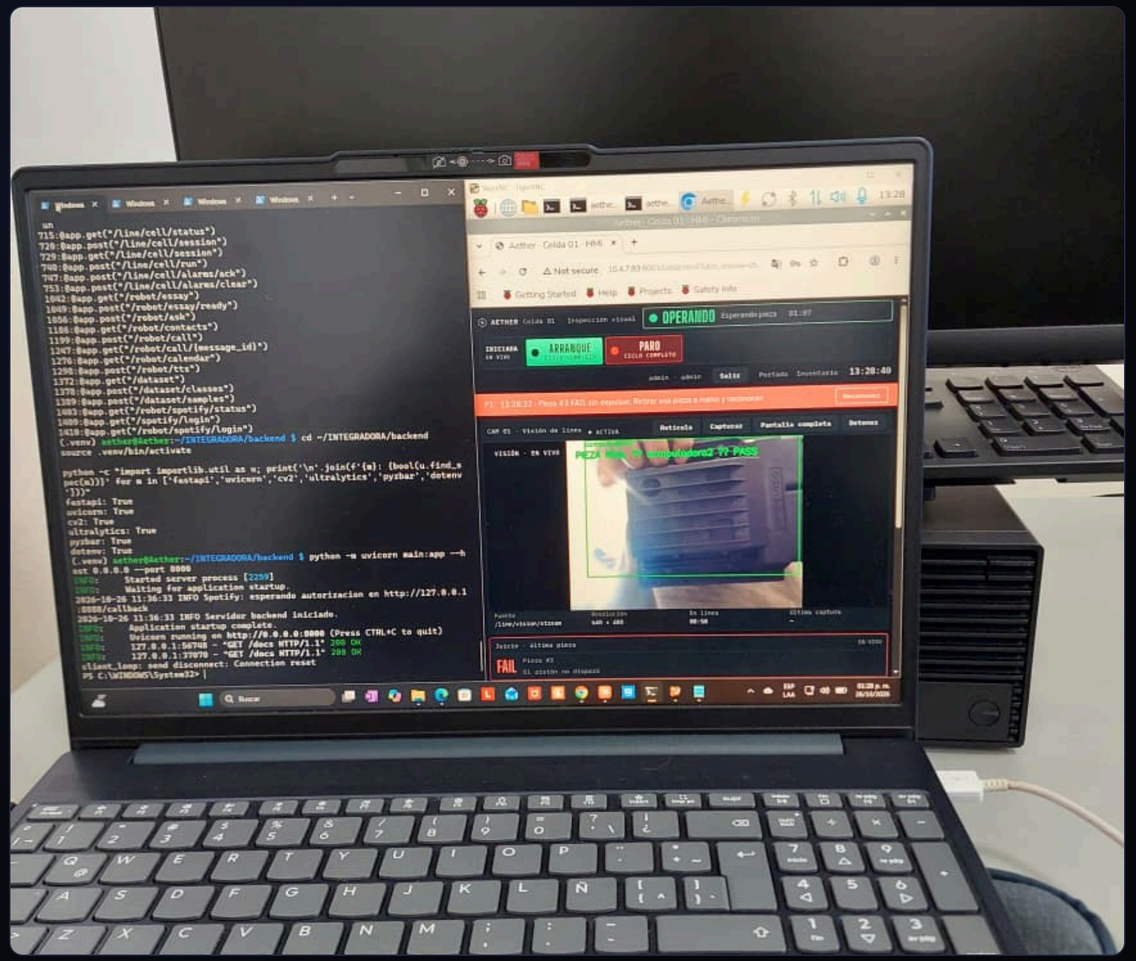
LOGO! ladder program

CONSTRUCTION · STAGE 4

Vision on the Raspberry Pi



Raspberry Pi 4 start-up and first tests



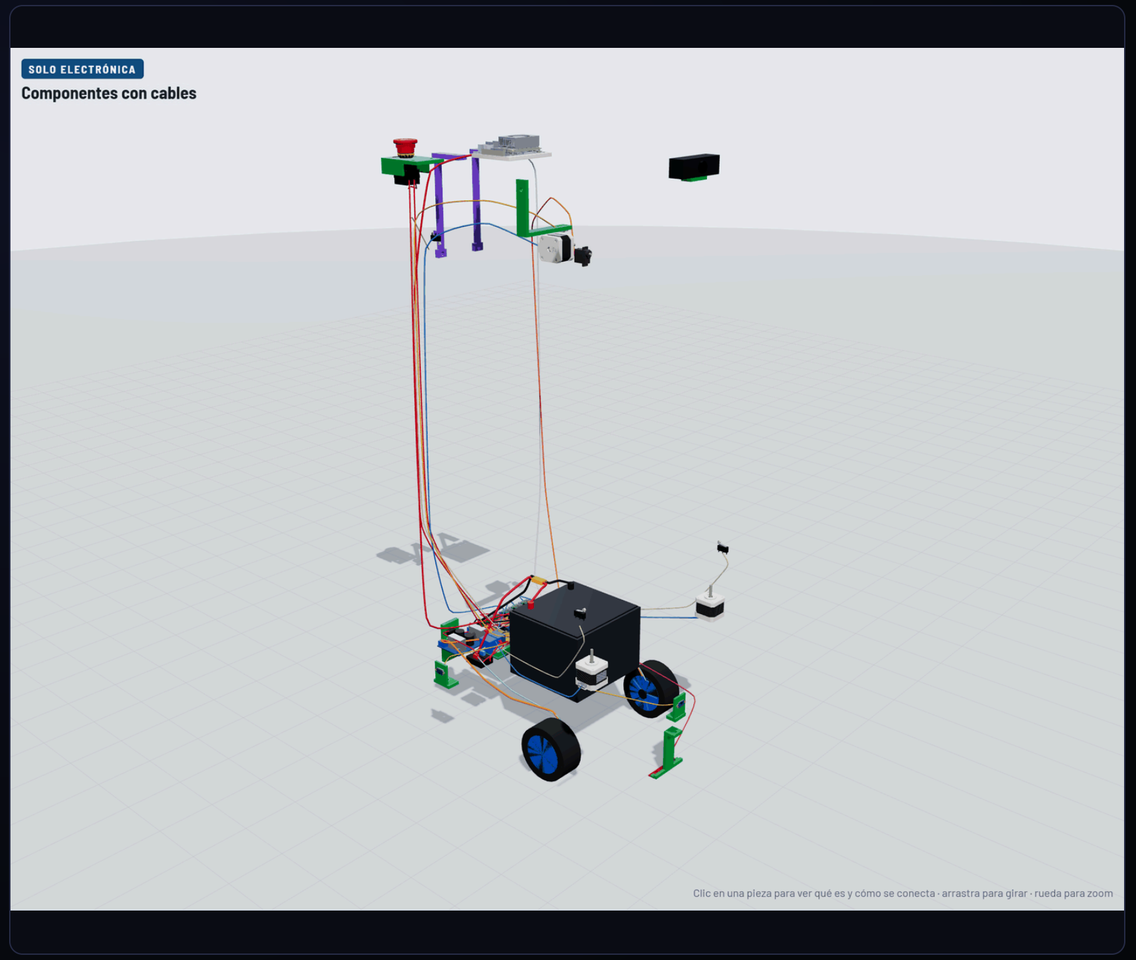
Vision tested with real parts

CONSTRUCTION · STAGE 5

The autonomous cart



Structure: elevator tower, battery and electronics



Electronics and wiring model

MAINTENANCE

Keeping the system running

Daily

Test the emergency stop. Clean camera lenses. Check air pressure and leaks.

Weekly

Check belt tension, fixtures and wheels. Charge and inspect the battery.

Monthly

Tighten terminals and profiles. Check fuses. Back up data and re-validate vision.

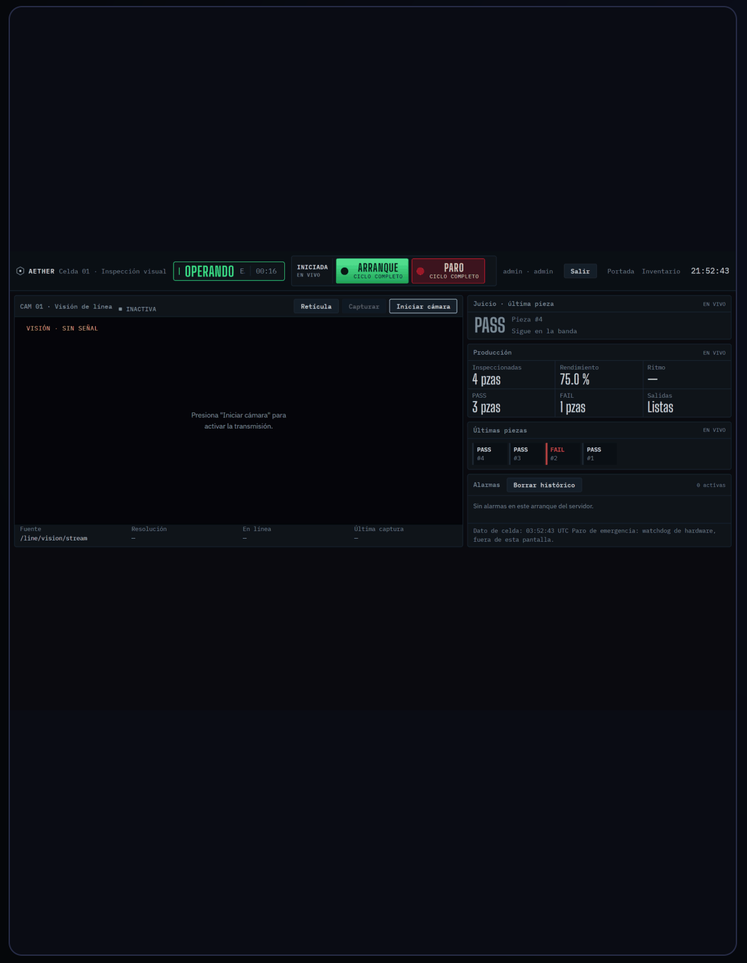
SAFETY AND STANDARDS

Stopping never depends on the network

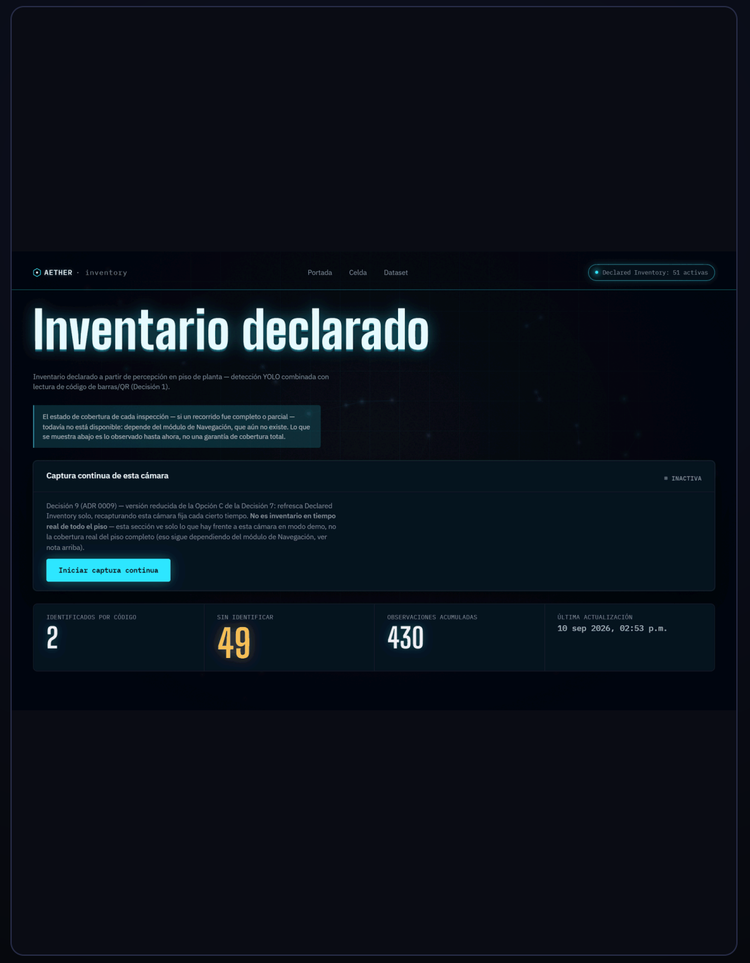
STANDARD	HOW IT APPLIES
ISO 12100	Emergency stop, guards, separated power branches
ISO 13849-1	The stop cuts power by hardware relay, not by software
NOM-004-STPS	Accessible emergency stop, moving parts identified
NOM-001-SEDE	Fuses per branch, wire gauges, 24 V apart from 5 V
ISO 3691-4	Sensor braking under 40 cm, speed limited to 0.35 m/s
ISA-95 / 101 / 18.2	Plant layers, operator screens, alarm management

INTERFACE

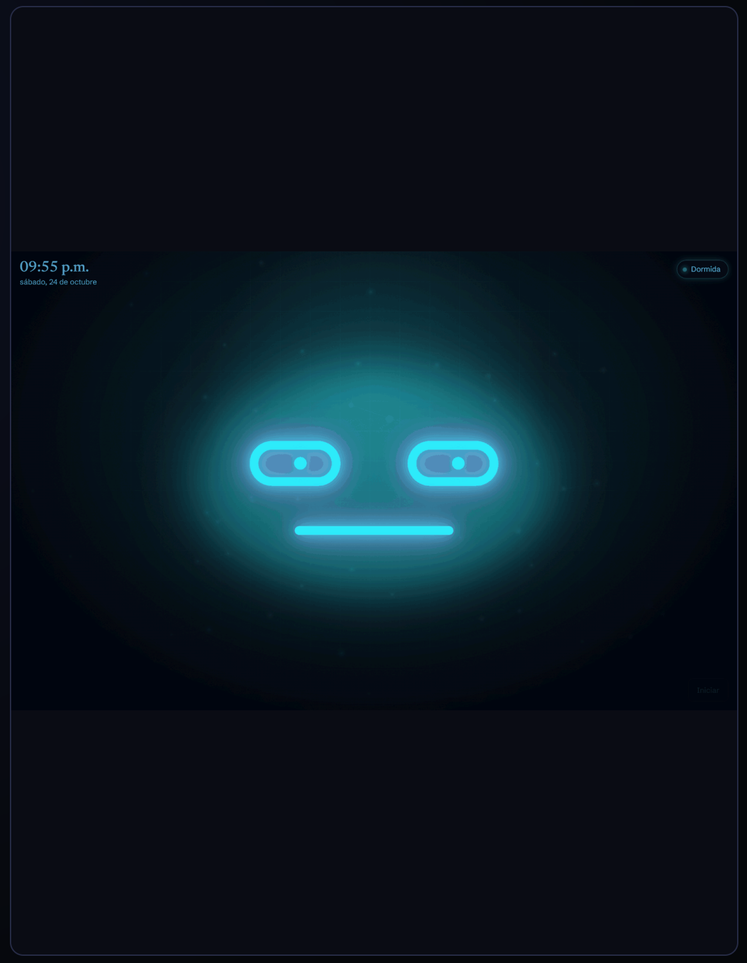
One interface for the whole line



Cell operator screen



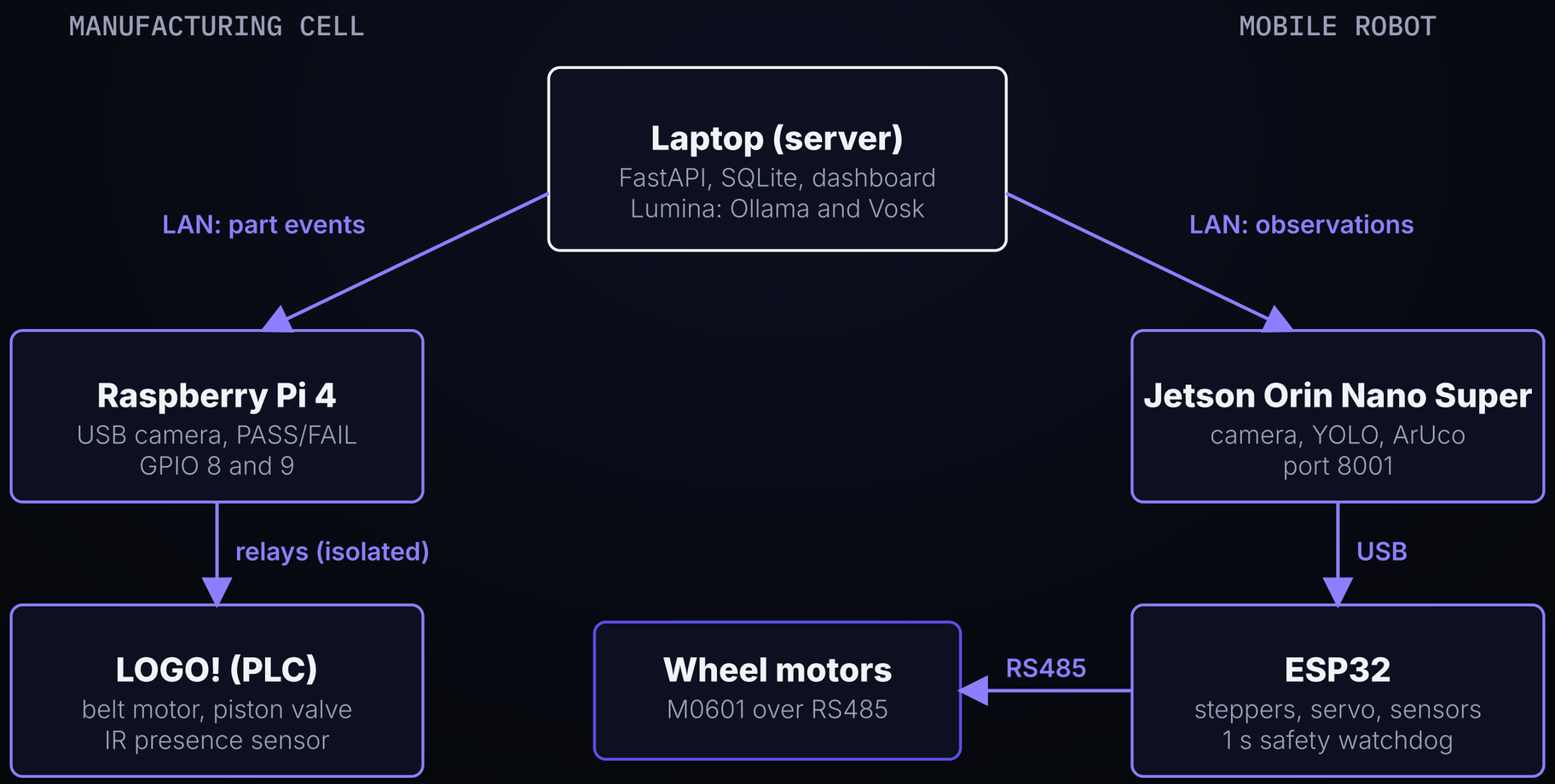
Declared inventory



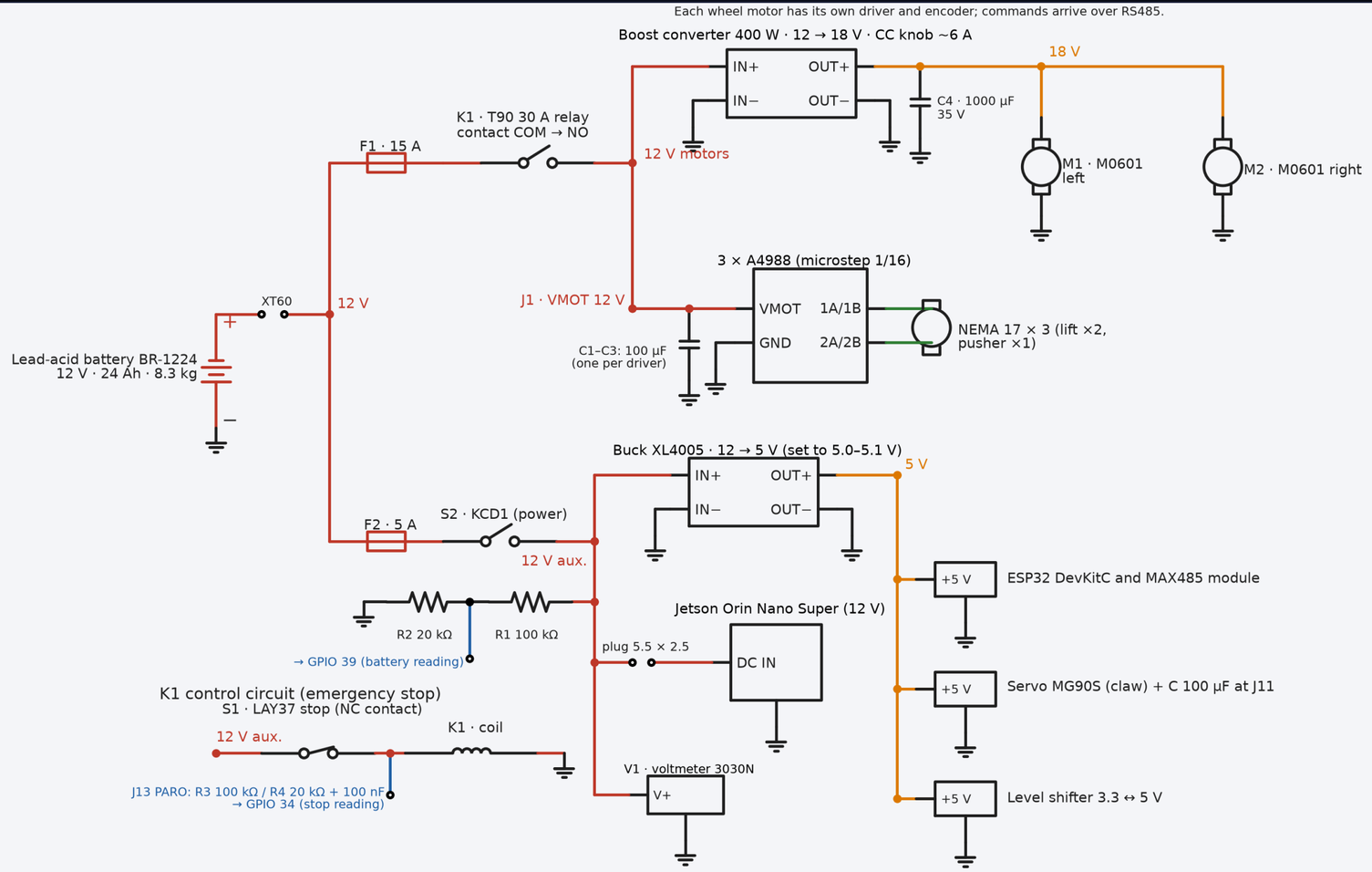
Lumina, the voice assistant

INTERCONNECTIONS

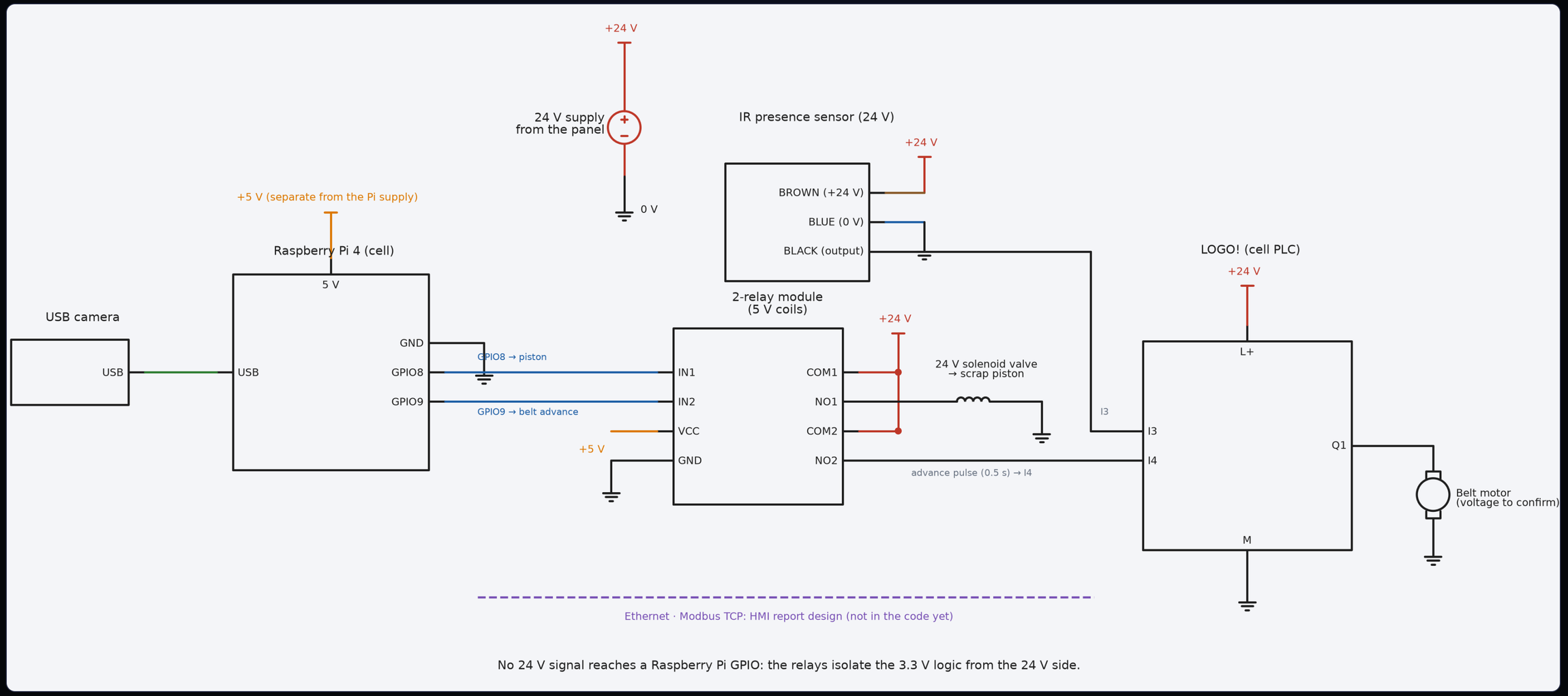
How the devices talk



Cart: power and emergency stop



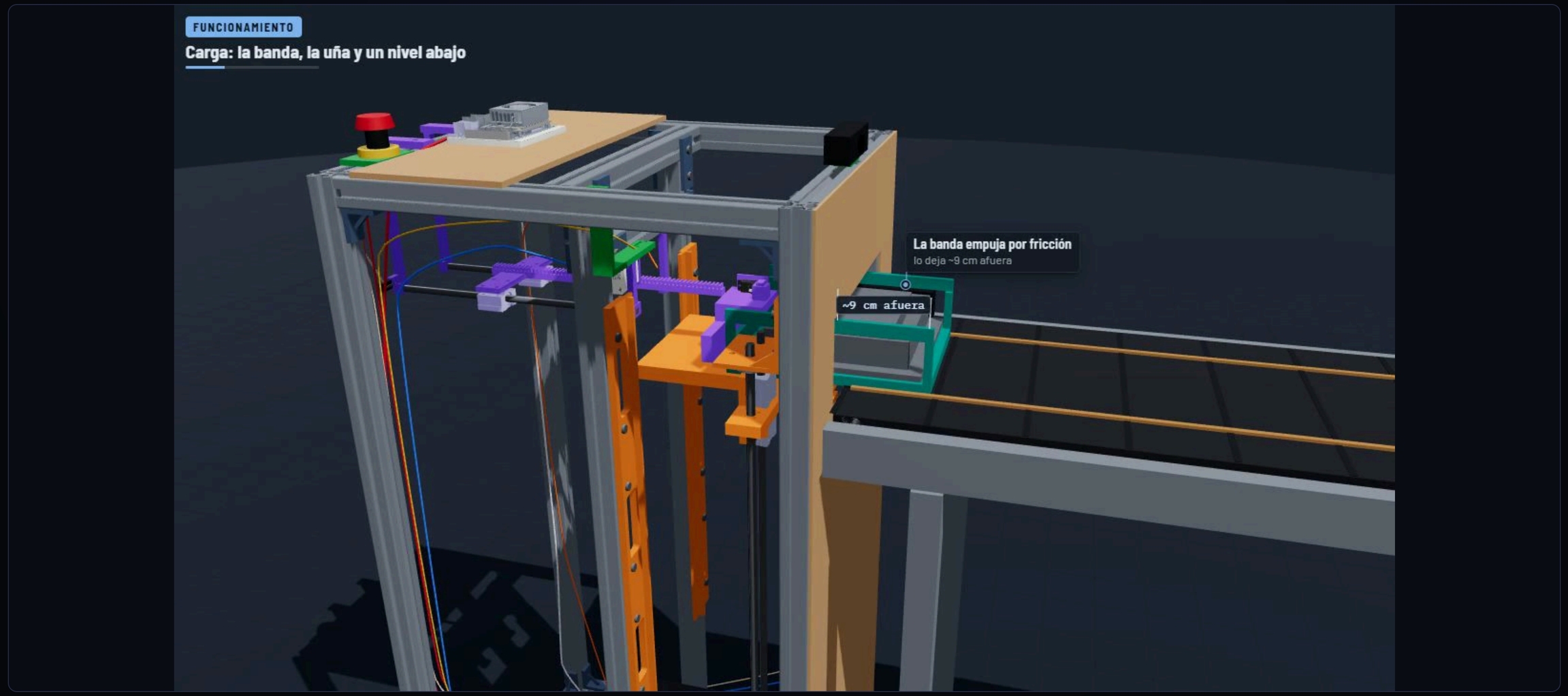
Cell: Raspberry Pi, relays and LOGO!



No 24 V signal reaches a Raspberry Pi GPIO: the relays isolate the 3.3 V logic from the 24 V side.

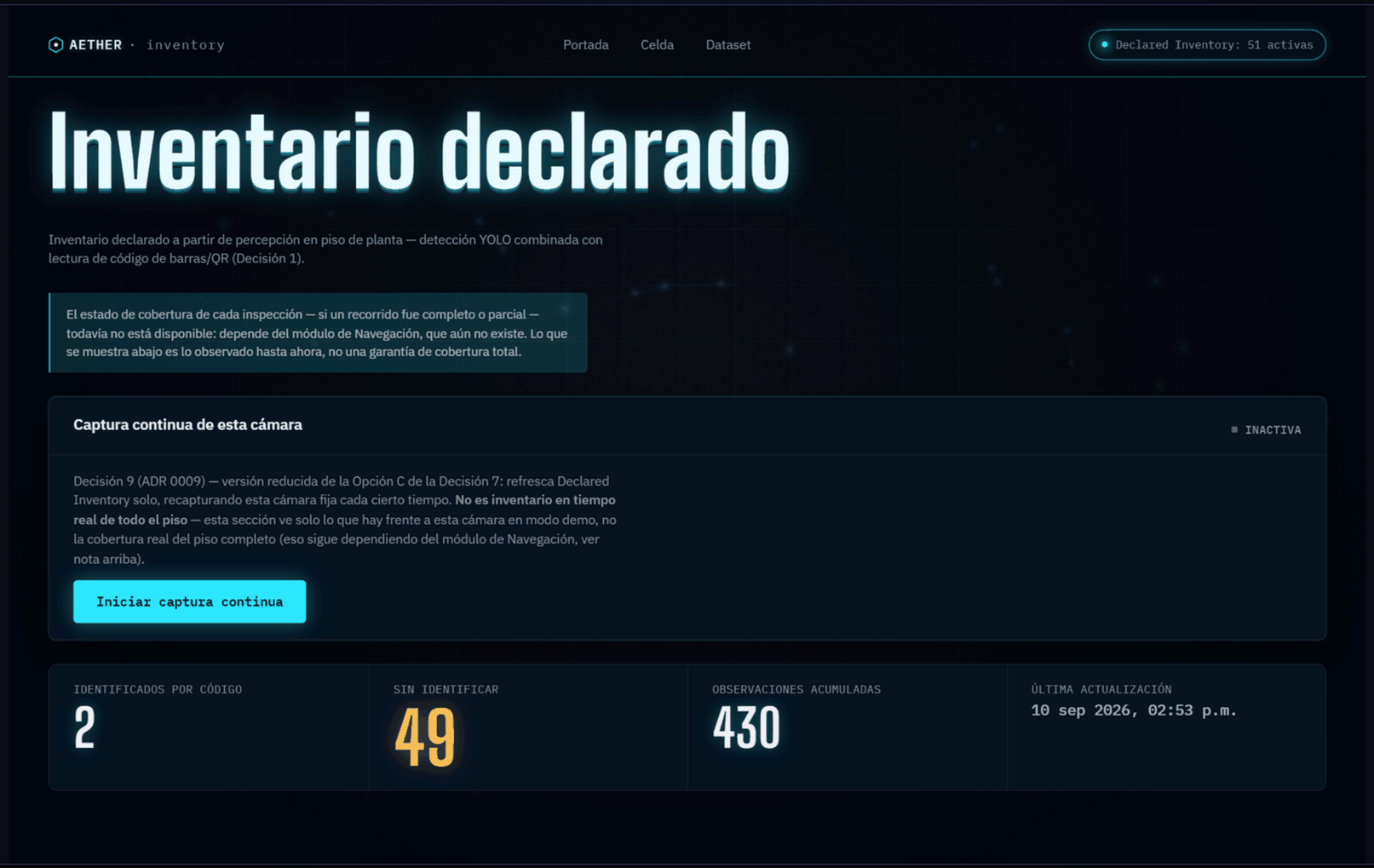
SIMULATION

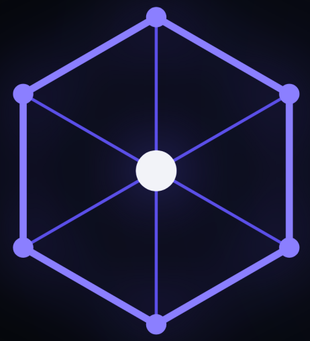
Simulated before it is built



REAL-TIME DEMO

Live: declared inventory from the server





Thank you.

AETHER · CHIHUAHUA, MEXICO